Web TECHNOLOGIES

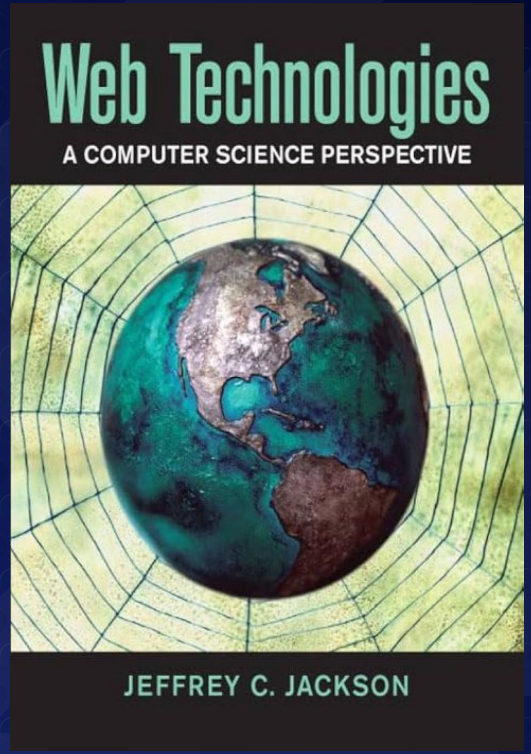
Course Objectives

1. To understand what is web technology
2. To learn why web technology is needed
3. To Develop web applications

Course Outcomes

1. Construct a basic website using HTML and Cascading Style Sheets
2. Build dynamic web page with validation using Java Script objects and by applying different event handling mechanisms
3. Construct simple web pages in PHP
4. Develop interactive web applications

Reference Book



Reference Websites

1. W3schools
2. Javatpoint

Course Outline

Web Essentials

Clients, Servers and Communication - The Internet - World wide web - HTTP vs HTTPS- Web Clients - Web Servers

HTML

DOM- Tables - Lists - Image - HTML control elements - Drag and Drop - Audio - Video controls

CSS

Inline, embedded and external style sheets - Rule cascading - Inheritance - Backgrounds - Border Images - Colors - Shadows - Text - Transformations - Transitions - Animations.

FRAMEWORK

Bootstrap Framework

JavaScript

Will be provided later...

PhP

Will be provided later...

Web Essentials

WEB TECHNOLOGY

What is Web Technology?

Web Technology refers to the tools and techniques used for creating, managing, and maintaining websites and web applications.

About building and delivering content and services over the Internet

COMPONENTS

Web Servers

Powerful computers that store and serve web content to users when requested.

Web Browsers

Primary interface through which users access the internet.

HTML

Markup language used to structure and present content on the web.

CSS

Visual presentation of web content

JavaScript

Dynamic programming language that adds interactivity and functionality to web pages

PHP

General-purpose scripting language geared towards web development

ESSENTIAL TOOLS

Notepad++

VSCode

Xampp

CLIENTS

What is a client?

The software that resides on the remote machine, communicates with the server, fetches the information, processes it, and then displays it on the remote machine is called the client.

-What is web client?

SERVERS

What is a server?

The software that distributes the information and the machine where the information and software reside is called the server

With the Internet, it's possible to access almost any information, communicate with anyone else in the world, and do much more

-What is web server?

What is a web client?

The client side (user side) of the Web.

A Web client typically refers to the Web browser in the user's machine or mobile device

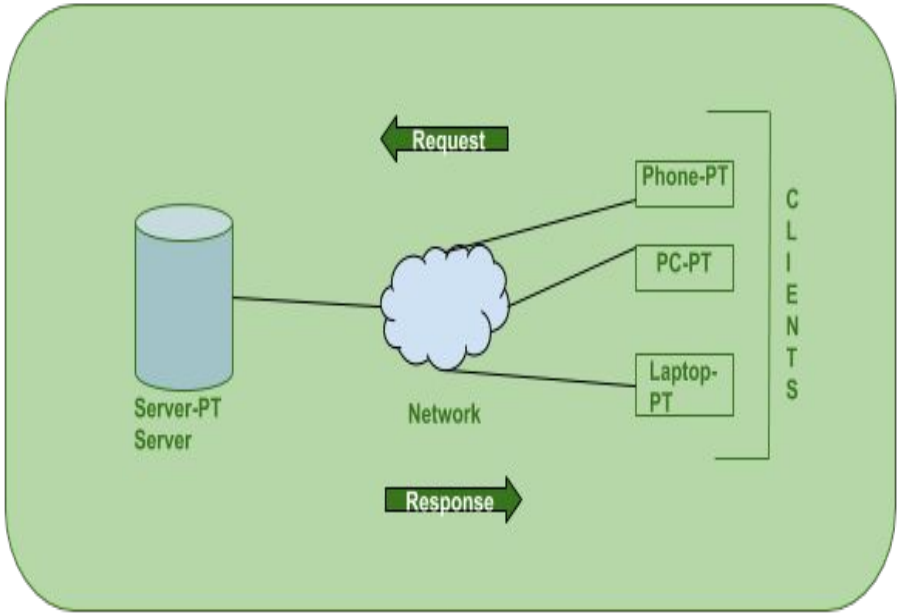
SERVERS

What is a web server?

A web server is software and hardware that uses HTTP and other protocols to respond to client requests made over the World Wide Web.

Display website content through storing, processing and delivering webpages to users.

CLIENT SERVER COMMUICATION



Clients and servers exchange messages in a request response messaging pattern.
The client sends a request, and the server returns a response.

To communicate, the computers must have a common language, and they must follow rules so that both the client and the server know what to expect.

CLIENT SERVER COMMUICATION

User enters the URL(Uniform Resource Locator) of the website or file. The Browser then requests the DNS(DOMAIN NAME SYSTEM) Server.

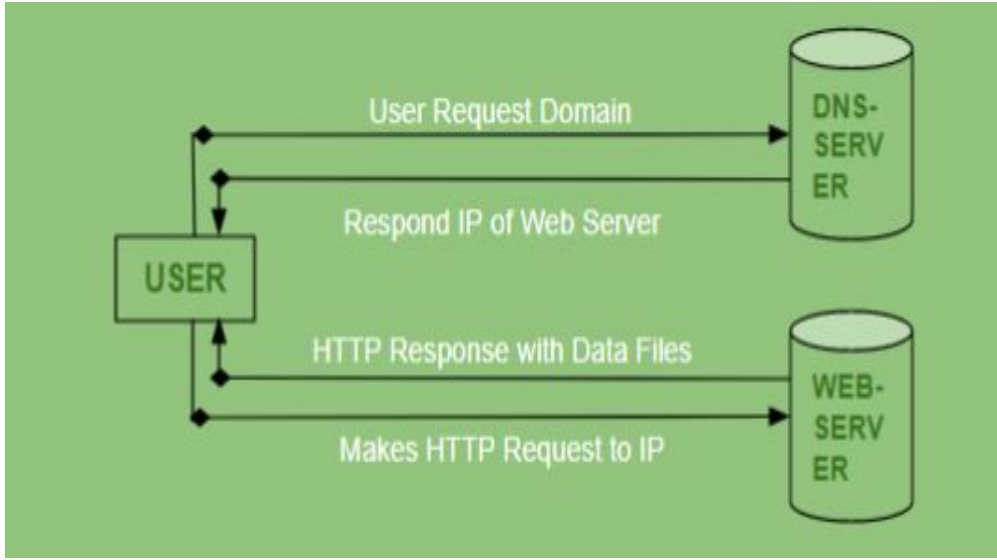
DNS Server lookup for the address of the WEB Server.

DNS Server responds with the IP address of the WEB Server.

Browser sends over an HTTP/HTTPS request to WEB Server's IP.

Server sends over the necessary files of the website.

Browser then renders the files and the website is displayed.



The Internet

- It is the largest network in the world that connects hundreds of thousands of individual networks all over the world.
- The popular term for the Internet is the "***information highway***".
- With the Internet, it's possible to access almost any information, communicate with anyone else in the world, and do much more.
- Rather than moving through geographical space, it moves your ideas and information through ***cyberspace – the space of electronic movement of ideas and information***.

Uses of The Internet

WEB

- The ***Web (World Wide Web)*** consists of information organized into Web pages containing text and graphic images.
- It contains ***hypertext links***, or ***highlighted keywords*** and images that lead to related information.
- ***A collection of linked Web pages that has a common theme or focus is called a Web site***.
- The main page that all of the pages on a particular Web site are organized around and link back to is called the site's home page.

Hypertext Mark-up Language (HTML)

- HTML stands for Hyper Text Markup Language
- HTML is the standard markup language for creating Web pages and describes the structure of a Web page
- HTML consists of a series of elements (DOM)
- HTML is made of tags (empty tag, container tag)

URL - OVERVIEW

A URL (Uniform Resource Locator) is a form of URI and standardized naming convention for addressing documents accessible over the ***Internet***.
Below is additional information about each of the sections of the http URL for this page.

https://www.lus.ac.bd/academic/department-of-cse

https:// - Protocol
www - Subdomain
lus.ac.bd - Domain and domain sufix
academic - Directories
department-of-cse - Web Page

URL – http or https

- The "***http***" stands for ***Hypertext Transfer Protocol***.
- It lets the browser to know which type of ***protocol*** it is going to use to access the information specified in the domain.
- An "***https***" protocol is short for "***Hypertext Transfer Protocol Secure***" and indicates that information transmitted over HTTP is ***encrypted and secure***.
- After the http or https is the ***colon (:)*** and ***two forward slashes (//)*** that separate the protocol from the remainder of the URL.
- A URL is not explicit to an HTTP or HTTPS addresses; FTP, TFTP, Telnet, and other addresses are also considered URLs and may not follow the same syntax as our example.

URL – *www*

- www stands for ***World Wide Web*** and is used to distinguish the content.
- This portion of the URL is not required and many times can be left out.
- For example, typing ***https://www.lus.ac.bd/academic/department-of-cse*** would still get you to the website.
- This portion of the address can also be substituted for an important sub page known as a ***subdomain.***

URL – *lus.ac.bd*

- ***lus.ac.bd*** is the domain name for the website.
- The last portion of the domain is known as the ***domain suffix***, or TLD.
- It is used to identify the type or location of the website.
- ***.bd is short for Bangladesh***
- There are several domain suffixes available. To get a domain, you would register the name through a domain registrar.

URL – academic

- ***Academics -*** the directories where the web page is on the server.
- To find the file on the server, it would be in the ***/public_html/academics directory***.
- With most servers, the public_html directory is the default directory containing the HTML files.

URL

- ***Is an IP address the same as a URL or web address?***

- No.
- An IP address is a unique number that's assigned to each device on a network.
- On the World Wide Web, a domain name is assigned a unique IP address.
- When typed (***lus.ac.bd***), DNS translates the domain name into an IP address that routers use to find the web server.
- A domain name is used instead of an IP address because it's easier for humans to remember.

Hyper Text Transfer Protocol

The Hypertext Transfer Protocol (HTTP) is the foundation of the World Wide Web, and is used to load webpages using hypertext links.

HYPERTEXT TRANSFER PROTOCOL

HTTP is a ***Connectionless protocol*** which can deliver any sort of data

The basic structure of HTTP communication follows ***request - response model***

HYPERTEXT TRANSFER PROTOCOL

The HTTP's basic structure includes two message types.

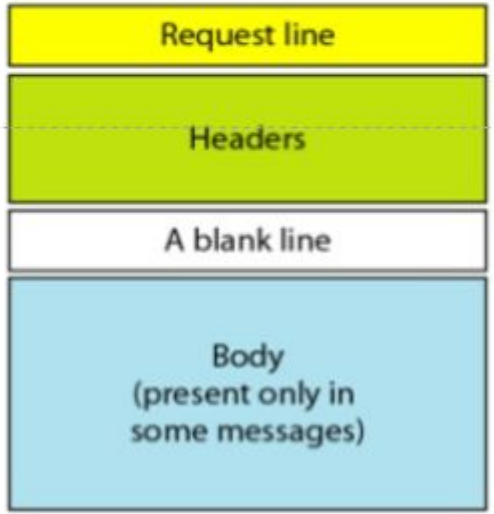
The Request Message
The Response Message

HTTP - REQUEST MESSAGE

HTTP requests are ***messages sent by the client to initiate an action on the server***.

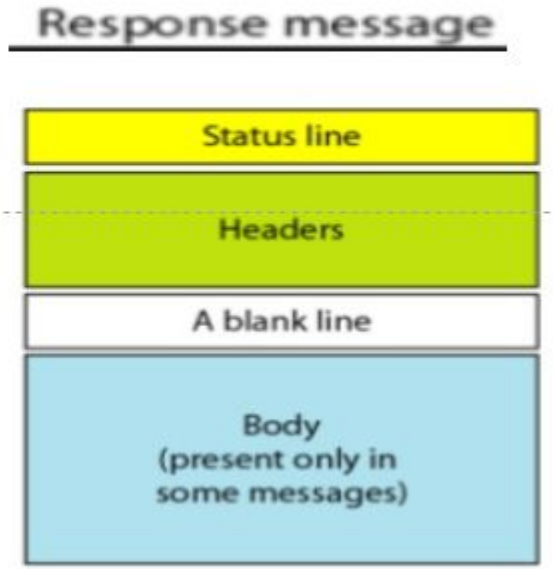
Every HTTP request message has the same basic structure:

- ***Request / Start line***
- ***Header field(s)*** (one or more)
- ***Blank line***
- ***Message body*** (optional)



RESPONSE MESSAGE

- Each Request has a Response.
- Every HTTP Response message consists of the follwing:
  - ***Status line***
  - ***Header field(s) (one or more)***
  - ***Blank line***
  - ***Message body (optional)***



- Hyper Text Mark-up Language
- Skeleton of all web pages.
- Provides structure to the content appearing on a website
- Created by Tim Berners-Lee in 1991

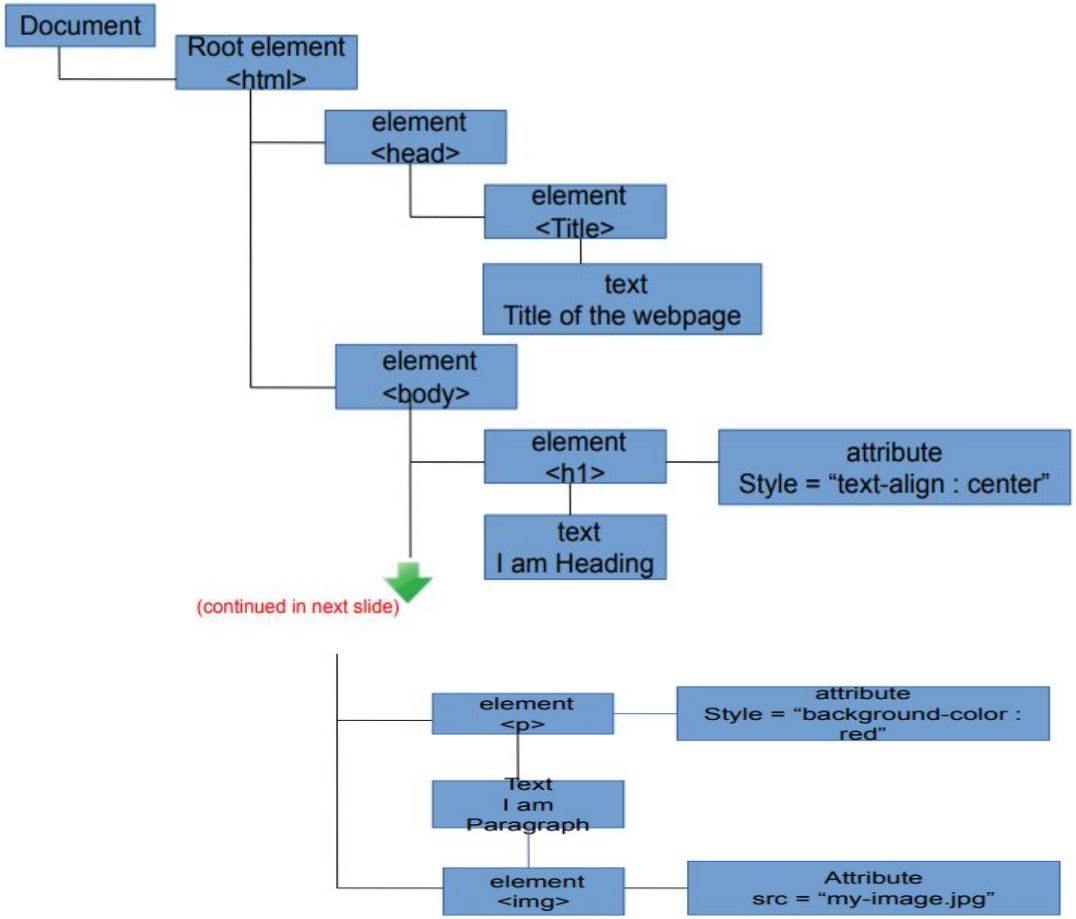
HTML Version	Year
HTML 1.0	1991
HTML 2.0	1995
HTML 3.2	1997
HTML 4.01	1999
XHTML	2000
HTML 5	2014

DOM

DOM – Document Object Model

- When a web Page is loaded in the browser, the browser creates a DOM of the Page
- The HTML DOM Model is Constructed as a tree of objects. It helps to define HTML elements as Object
- Dom provides an API(Programming Interface) for JS, using which we can manipulate the Dom, helping create dynamic HTML

DOM – Document Object Model

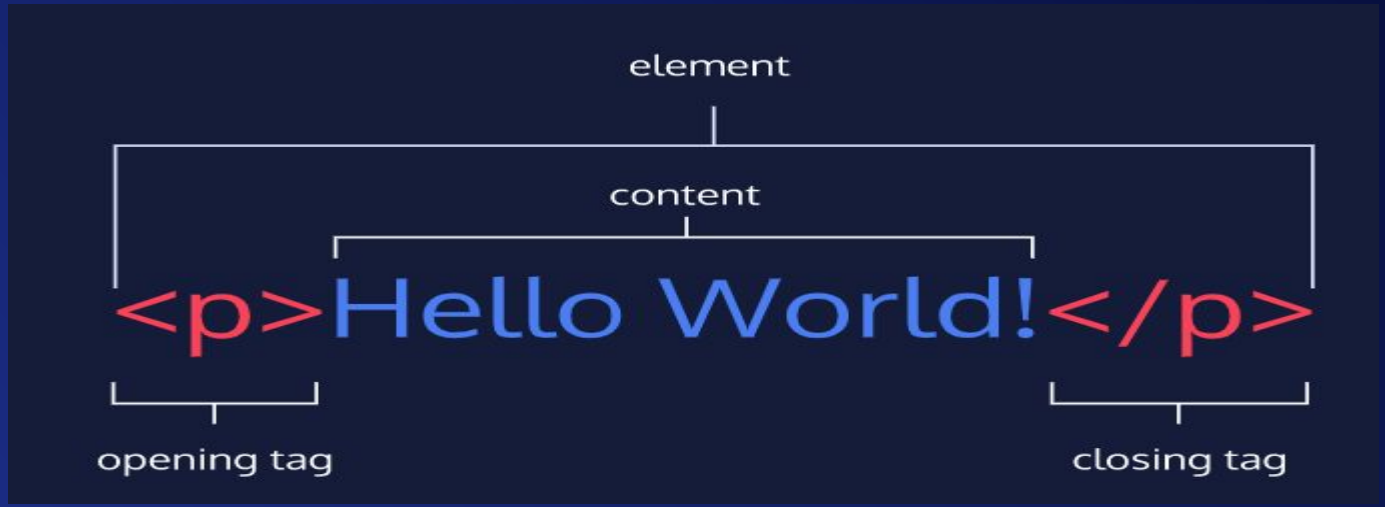


HTML - Structure

- HTML is composed of elements that structure the webpage and define its content.
- The ***elements are the building blocks*** of any HTML page and are represented by ***tags***.
- Tags are one of the most important part in an HTML Document.
- HTML uses some ***predefined tags*** which tells the browser about content display property, that is how to display a particular given content.

HTML - Structure

- For Example, to create a paragraph, one must use the paragraph tags***( )*** and to insert an image one must use the img tags***(<img />)***



HTML - Structure

- An ***HTML Document*** is mainly divided into two parts:
  - ***HEAD***: This contains the information about the HTML document. For Example, Title of the page, version of HTML, Meta Data etc.
  - ***BODY***: This contains everything you want to display on the Web Page.

```
<!DOCTYPE html>
    <html>

        <head>
            <title> Title here </title>
        </head>

        <body>
            Web page content goes here.
        </body>

    </html>
```

HTML - Structure

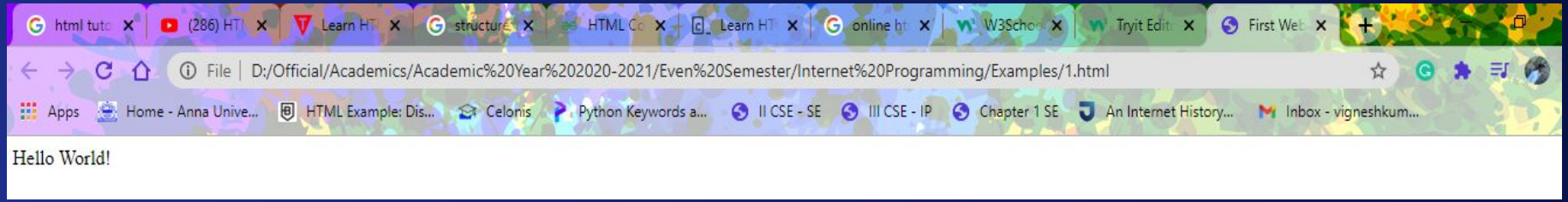
- Every Webpage must contain the above code.
- ***<!DOCTYPE html>:*** This tag is used to tells the ***HTML version.*** This currently tells that the version is ***HTML 5***.
- **<html>:** This is called ***HTML root element*** and used to wrap all the code.
- **<head>:** Head tag contains ***metadata, title, page CSS*** etc.
- **<body>:** Body tag is used to enclosed all the data which a web page has from texts to links.
- All of the content that you see rendered in the browser is contained within this element.

HTML – First Web Page

- ***Example:***

```
<html>
    <head>
        <title>
            First Web Page
        </title>
    <meta charset="UTF-8">
    </head>

    <body>
        Hello World!
    </body>
</html>
```

- ***Output:***



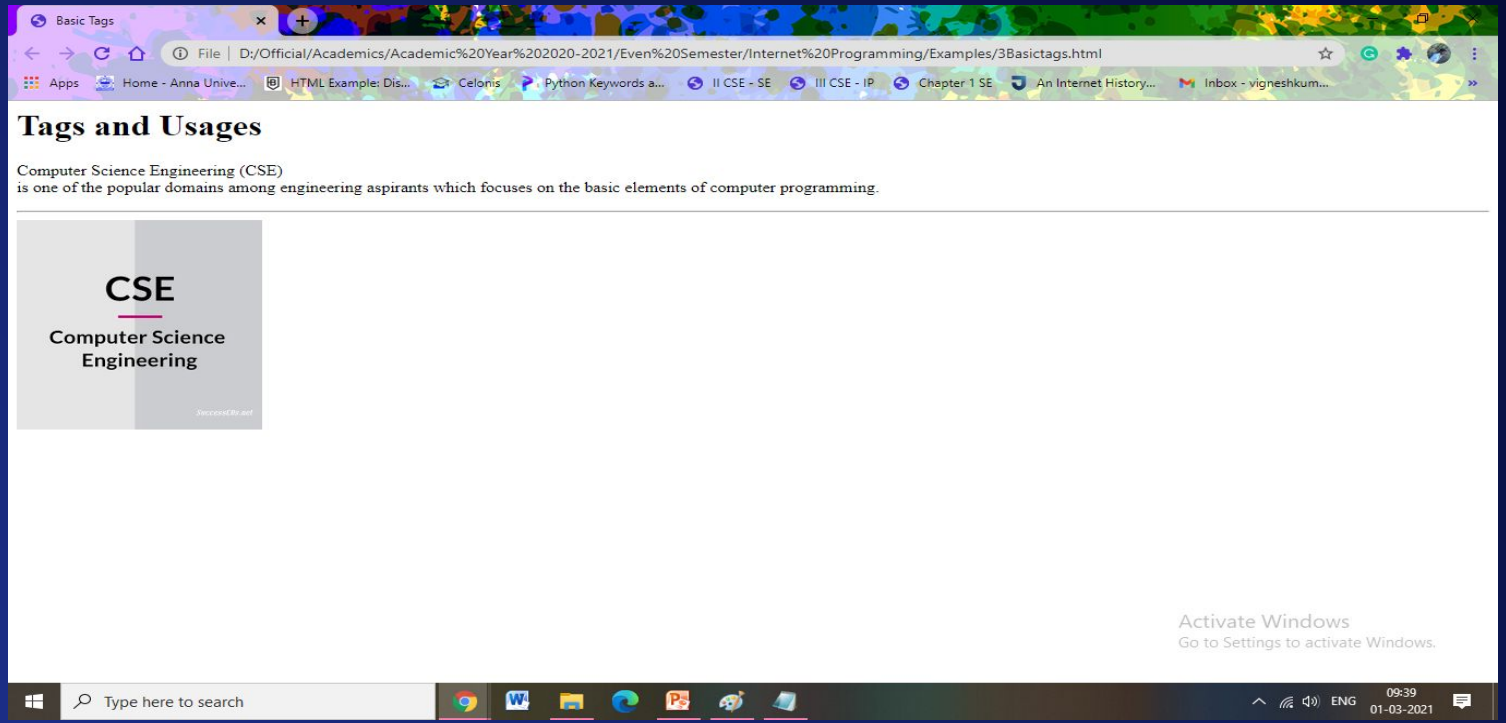
Basic HTML Tags & Attributes

- The following tags help in organization and basic formatting of elements in our script or web pages.
  - ***HTML Paragraph *** - to write paragraph statements in a webpage.
  - ***Break </br> -*** to break line and acts as a carriage return.
  - ***Horizontal Rule <hr>*** - to break the page into various parts, creating horizontal margins
  - ***HTML Images <img src="source_of_image"***> - to insert an image into our web page
  - ***HTML Attributes*** - to provide extra or additional information about an element
  - ***HTML Comments*** - for inserting comments in the HTML code

Basic HTML Tags & Attributes

```
<html>
<head>
<!-- Example for Basic Tags -->
        <title>Basic Tags</title>
    <meta charset="UTF-8">
</head>
<body>
        <h1>Tags and Usages</h1>
        <p>
        Computer Science Engineering (CSE) <br>is one of  the popular domains among engineering
aspirants which focuses on the basic elements
    of computer programming.<br> </p>
        <hr>
<img src="download.jpg">
</body>
</html>
```

Basic HTML Tags & Attributes



Introduction to HTML5

History

1991	HTML first published
1995	HTML 2.0
1997	HTML 3.2
1999	HTML 4.01
2000	XHTML 1.0
2002-2009	XHTML 2.0
2012	HTML5
2014	HTML5.1

After HTML 4.01 was released, focus shifted to XHTML and its stricter standards.

XHTML 2.0 had very strict standards but was abandoned in 2009 in favor of HTML5.

HTML5 is much more tolerant and can handle markup from all the prior versions. A working draft was released in 2012 and it is scheduled to be finalized by the end of 2014.

HTML continues to evolve, with ongoing developments and proposed features aiming to enhance its capabilities and address emerging needs. So far released versions are HTML5.2 in 2017, HTML5.3 in 2018.

FEATURES

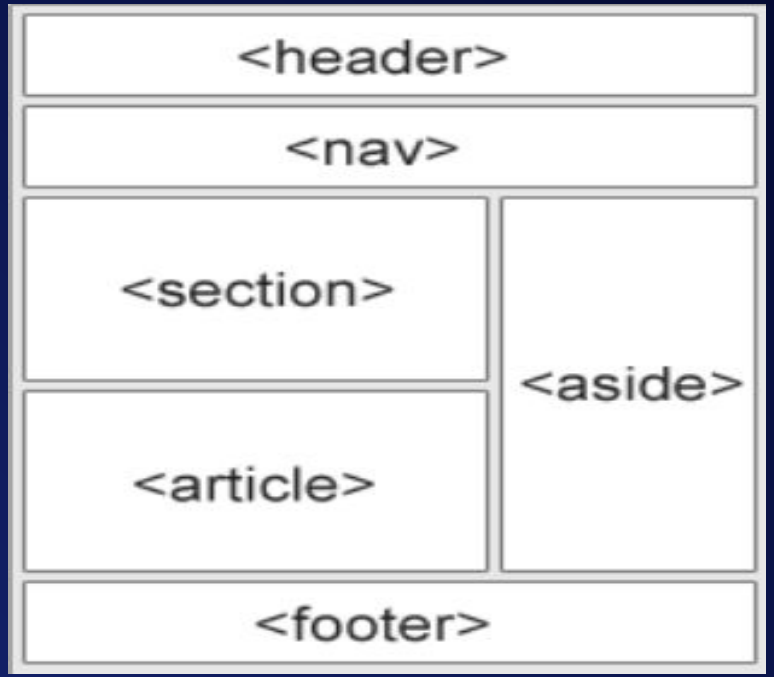
- Built-in ***audio and video support*** (without plugins)
- Enhanced ***form controls and attributes***
- ***The Canvas*** (a way to draw directly on a web page)
- ***Drag and Drop*** functionality
- ***Support for CSS3*** (the newer and more powerful version of CSS)
- More advanced features for web developers, such as ***data storage and offline applications***.

HTML5

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="utf-8">
    <title>My First HTML5 Page</title>
    <link rel="stylesheet" href="style.css">
  </head>
  <body>
    <p>HTML5 is fun!</p>
  </body>
</html>
```

HTML 5 - Semantics

- ***Semantic Tags:*** A semantic element clearly describes its meaning to both the browser and the developer.
- HTML5 introduces many semantic elements which make the code easier to write and understand for the developer as well as instructs the browser on how to treat them.
- A Semantic Web ***"allows data to be shared and reused across applications, enterprises, and communities."***



HTML - Lists

- A list is a ***record of short pieces of information***, such as people's names, usually written or printed with a single thing on each line and ordered in a way that makes a particular thing easy to find.
- HTML offers three ways for specifying lists of information. All lists must contain one or more list elements.
  - ***An unordered list - ul***
  - ***An ordered list – ol***
  - ***dl,dt,dd***

An Unordered Lists

- An unordered list starts with the ***"ul"*** tag.
- Each list item starts with the ***"li"*** tag.
- The list items are marked with bullets i.e. small black circles by default.
- The HTML Unordered List has various List Item Markers:-
  - ***Disc***
  - ***Circle***
  - ***Square***

```
<ul style="list-style-type:disc/circle/square">
```

An Ordered Lists

- An ordered list starts with the ***"ol"*** tag.
- Each list item starts with the ***"li"*** tag.
- The list items are marked with numbers by default.
- The HTML Ordered List has various List Item Markers:
  - ***Items numbered with numbers***
  - ***Items numbered with uppercase letters***
  - ***Items numbered with lowercase letters***
  - ***Items numbered with uppercase roman numbers***
  - ***Items numbered with lowercase roman numbers***

<ol type="1" / "A" / "a" / "I" / "i">

Lists Example

Code

```
<!DOCTYPE html>
<html>
<body>

<h2>Unordred Lists</h2>

<ul>
<li>Pulsar 150</li>
<li>Pulsar 160</li>
<li>Pulsar 200</li>
<li>Pulsar 220</li>
</ul>

<h2>Ordred Lists</h2>

<ol>
<li>Pulsar 150</li>
<li>Pulsar 160</li>
<li>Pulsar 200</li>
<li>Pulsar 220</li>
</ol>
</body>
</html>
```

Output

HTML 5 - Header

- The ***<header>*** tag in HTML is used to define the header for a document or a section.
- ***Syntax:***

```
<header> ...</header>
```

- The header tag contains information related to the title and heading of the related content.
- The browsers which supports <header> tag are : ***Google Chrome 6.0, Internet Explorer 9.0, Firefox 4.0, Opera 11.1, Safari 5.0***

Header Example

```
<head>
<title>Header Tag</title>
</head>
<body>
<header>
<h1>KIT</h1>
<p>
<a href="https://karpagamtech.ac.in/about-us/">About Us</a> |
<a href="https://karpagamtech.ac.in/academics/computer-science-engineering/"> CSE</a> |
<a href="https://karpagamtech.ac.in/academics/electronics-and-communication-
engineering/"> ECE</a> |
<a href="https://karpagamtech.ac.in/academics/mechanical-engineering/"> Mech</a> |
<a href="https://karpagamtech.ac.in/academics/electrical-and-electronics-engineering/">
EEE</a> |
<a href="https://karpagamtech.ac.in/academics/information-technology/"> IT</a> |
<a href="https://karpagamtech.ac.in/contact-us/">Contact Us</a>
</p>
</header>
</body>
</html>
```

HTML 5 - Footer

- The ***<footer>*** element specifies a ***footer for a document or section.***
- A element should contain ***information about its containing element***.
- A footer typically contains the author of the document, copyright information, links to terms of use, contact information, etc.
- ***Syntax:***

  <footer> ... </footer>

Footer - Example

```
<!DOCTYPE html>
<html>
   <head>
      <title>footer Tag</title>
   </head>
   <body>
      <footer>
         <p>Copyrights Reserved: KIT</p>
         <p>Contact: <a href="https://karpagamtech.ac.in/contact-us/">
            karpagamtech.ac.in</a>.
         </p>
      </footer>
   </body>
</html>
```

HTML 5 - NAV

- The <nav> element defines a set of navigation links.
- Websites typically have sections dedicated to navigational links, which enables users to navigate the site.
- These links can be placed inside a nav tag.
- The links in the nav element may point to other webpages or to different sections of the same webpage.
- ***Syntax:***

  <nav> Links </nav>

HTML 5 - Aside

- The ***<aside>*** tag is used to ***describe the main object of the web page in a shorter way like a highlighter***.
- It basically identifies the content that is related to the primary content of the web page but does not constitute the main intent of the primary page.
- The <aside> tag contains mainly author information, links, related content, and so on.
- The <aside> tag makes it ***easy to design the page and it enhances the clarity of HTML document***.
- ***Syntax:***

```
<aside>
<h1>Contents...</h1>
</aside>
```

HTML 5 - Article

- The element specifies independent, self-contained content.
- An ***article*** should make sense on its own, and it should be possible to read it independently from the rest of the web site.
- Can be used for

  Forum Post
  Blog Post
  Newspaper Article

- ***Syntax***

```
<article>
Contents
</article>
```

HTML 5 - Article

```
<!DOCTYPE html>
<html>
<body>

<h1>The article element</h1>

<article>
  <h2>Google Chrome</h2>
  <p>Google Chrome is a web browser developed by Google, released in 2008. Chrome is the
world's most popular web browser today!</p>
</article>

<article>
  <h2>Mozilla Firefox</h2>
  <p>Mozilla Firefox is an open-source web browser developed by Mozilla. Firefox has
been the second most popular web browser since January, 2018.</p>
</article>

<article>
  <h2>Microsoft Edge</h2>
  <p>Microsoft Edge is a web browser developed by Microsoft, released in 2015. Microsoft
Edge replaced Internet Explorer.</p>
</article>

</body>
```

HTML 5 - FIGURE AND FIGCAPTION

- These are used to add an image in a web page with small description.

```
<!DOCTYPE html>
<html>
<head>
    <title>Figcaption Tag</title>
    </head>
<body>
    <h2>Figure and Fig Caption</h2>
    <figure>
        <img src="logo.png" alt="gfg">
        <figcaption style="color:red">Karpagam Logo</figcaption>
    </figure>
</body>
</html>
```

HTML 5 - Control Elements

- ***HTML Control Elements*** are responsible to accept the ***User input in a specified manner.***
- These elements can be grouped together inside a form to collect data from a User in a User-friendly manner.

HTML 5 - Input Text Control

- Input text controls are used to ***collect User data as a free text***.
- On the web page, it will form a ***rectangle box*** where Users can input the data.
- There are different types of input text controls that can be used in the HTML forms.
  - ***Single line Input Text Control***
  - ***Multi-line Input Text Control***
  - ***Password Input Control***

Single line Input Text Control

- This allows the user to enter only a single line of data. A typical example of such input text controls is for entering the name, search box, city, etc.

```
<!DOCTYPE html>
<html>
<head>
    <title>Single Line Input Control</title>
    </head>
<body>
    <h2>Forms</h2>
    <form>
    <input type="text" name="Firstname">
    <br>
    <br>
    <input type="submit" name="submit">
    </form>
</body>
</html>
```

Forms

Vignesh

Submit

Multi-line Input Text Control

- This input control type allows the user to enter data of multiple lines.
- Typical usage of such input controls is for comments, addresses, description and so on.

```
<!DOCTYPE html>
<html>
<head>
    <title>Single Line Input Control</title>
    </head>
<body>
    <h2>Forms</h2>
    <form>
    <textarea rows="5" cols="60" name="description">
Enter the details....
    </textarea>
    <br>
    <br>
    <input type="submit" name="submit">
    </form>
</body>
</html>
```

Forms

Enter the details....

Submit

Password Input Control

- This is typically used for the password field. This works in the same way as the input text field but the text gets masked for safety purposes.

```
<!DOCTYPE html>
<html>
<head>
    <title>Single Line Input Control</title>
    </head>
<body>
    <h2>Forms</h2>
Password
    <input type="Password"name="password">
    <br>
    <br>
    <input type="submit" name="submit">
    </form>
</body>
</html>
```

Forms

Password ••••••••••••••

Submit

Input Type Submit

- When input type is of submitting it performs the action defined in the form action and sends the form data to the server.

```
<!DOCTYPE html>
<html>
<head>
    <title>Single Line Input Control</title>
    </head>
<body>
    <h2>Forms</h2>
    <form action="/action.php"
    username:
    <input type="text" name="Username" value="ss">
    <br>
    <br>
Password
    <input type="Password"name="password">
    <br>
    <br>
    <input type="submit" name="submit">
    </form>
</body>
</html>
```

Forms

Password

Submit

Input Type Radio

- Radio buttons are used when you expect Users to fill data as a Boolean value or you expect only one input as true out of multiple options.

```
<!DOCTYPE html>
<html>
<head>
    <title>Radio Buttons</title>
    </head>
<body>
     <p>Employee Type</p>


    <input type="radio" name="type" value="regular" checked>Regular <br>
   <input type="radio" name="type" value="Temporary">Temporary <br>
   <input type="radio" name="type" value="other">Other <br>
    <input type="submit" name="submit">
    </form>
</body>
</html>
```

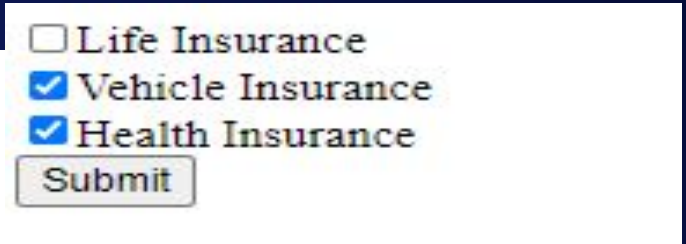
Employee Type

Regular
Temporary
Other
Submit

Input Type Checkbox

- A checkbox lets the User select whatever information is true in his case.
- It is a very convenient way of accepting the data when the possible input is already known.

```
<!DOCTYPE html>
<html>
<head>
    <title>Check Boxes</title>
    </head>
<body>
    <form>
    <input type="checkbox" name="LI" value="LI" checked>Life Insurance <br>
   <input type="checkbox" name="VI" value="VI">Vehicle Insurance <br>
   <input type="checkbox" name="HI" value="HI">Health Insurance <br>
    <input type="submit" name="submit">
    </form>
</body>
</html>
```



Input Type drop-down list

- The ***drop-down list*** enables the user to select ***one option out of multiple possible options***. This is a very User-friendly way to get the detail from the User as it provides an exhaustive list of possible options that helps the user to identify the option best suits him.

```
<!DOCTYPE html>
<html>
<head>
    <title>Check Boxes</title>
    </head>
<body>
    <form>
    <select name="location">
    <option value="Coimbatore">Coimbatore </option>
    <option value="Pollachi">Pollachi </option>
    <option value="Gobi">Gobi </option>
    <option value="Salem">Salem </option>
    </select>
    <br><br>
    <input type="submit" name="submit">
    </form>
</body>
</html>
```

Input Type Optgroup

- ***Optgroup*** works in a similar way as of the drop-down list, the only difference is that the ***optgroup lets you to logically group certain options under one umbrella***. It helps the user to quickly identify the relevant option with the help of the optgroup label.

```
<!DOCTYPE html>
<html>
<head>
    <title>Check Boxes</title>
    </head>
<body>
    <form>
    <select>
    <optgroup label="Tamilnadu">
    <option value="Coimbatore">Coimbatore </option>
    <option value="Pollachi">Pollachi </option>
    <option value="Gobi">Gobi </option>
    <option value="Salem">Salem </option>

    <optgroup label="Kerala">
    <option value="Coimbatore">Palakad </option>
    <option value="Pollachi">Pollachi </option>
    </select>
    <br><br>
    <input type="submit" name="submit">
    </form>
</body>
</html>
```

Fieldset

- ***Fieldset*** is another useful tag in the Html form which let the ***developer to logically group certain controls under one legend***, this help the developer to give User a clear instruction on what to expect in this section.

```
<!DOCTYPE html>
<html>
<head>

    </head>
<body>
    <form>

    <fieldset>
    <legend>Login:</legend>
    Username:<br>
    <input type="text" name="username">
    <br>
    Password:
    <br>
    <input type="password" name="password">
    <br><br>

    <input type="submit" name="submit">
    </fieldset>
    </form>
</body>
</html>
```

Login:

Username:

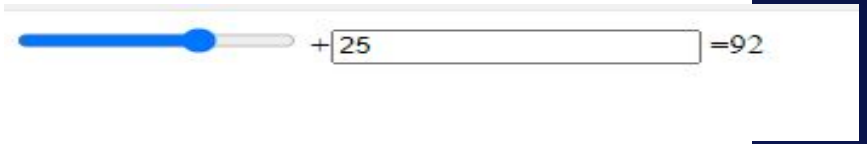
Password:

Submit

HTML output Tag

- It let you ***display the output*** of a calculation instantly. This is quite useful when the user needs to do ay calculation instantly and see the results.

```
<!DOCTYPE html>
<html>
<head>

    </head>
<body>
    <form oninput="x.value=parseInt(a.value)+parseInt(b.value)">
<input type="range" id="a" value="50">
+<input type="number" id="b" value="25">
=<output name="x" for="a b"></output>
</form>


    </form>
</body>
</html>
```



Input type Color

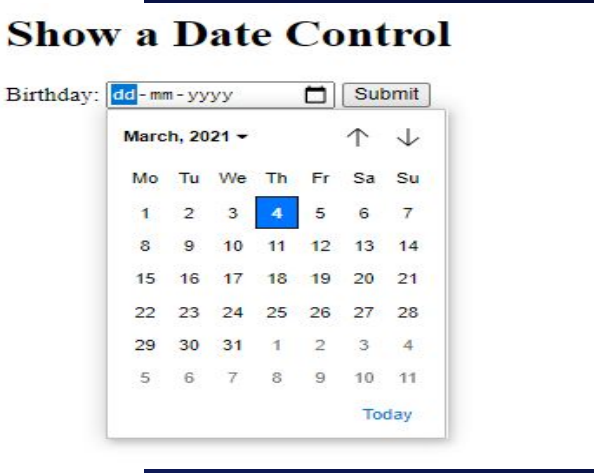
- It is often required in the form to just show the color instead of any text.
- It shows the color which you want to display in the form. Typical scenario's where it is being used is to show the status of a project or a phase.

```
<!DOCTYPE html>
<html>
<head>

    </head>
<body>
    <h1>Show a Color Picker</h1>

<form action="/action_page.php">
  <label for="favcolor">Select your favorite color:</label>
  <input type="color" id="favcolor" name="favcolor" value="#ff0000"><br><br>
  <input type="submit">
</form>

    </form>
</body>
</html>
```

Input type Date

- Input type date is commonly used where a date type field as input is expected by the User, it could be anything like a Date of Birth, Hiring date, termination date and so on.

```
<!DOCTYPE html>
<html>
<head>
</head>
<body>
    <h1>Show a Date Control</h1>
<form action="/action_page.php">
  <label for="birthday">Birthday:</label>
  <input type="date" id="birthday" name="birthday">
  <input type="submit">
</form>
 </body>
</html>
```



HTML 5 Tables

- The HTML table provides a ***way to derive or define the data such as text, images, links etc. in terms of rows and columns*** of cells.
- The HTML tables can be created by using ***<table>*** tag.
- The table can be created by using ***<th>, <td>, and <tr>*** tags.
  - The <th> tag defines a ***heading for the table***.
  - The <td> tag specifies the ***table data cells which is used to make the column.***
  - The <tr> tag specifies the ***table rows which is used to make a row.***
- ***The table data can be structured within <table>content of the table</table> with numerous table elements.***

Basic Table Usage

```html
<!DOCTYPE html>
<html>
<head>
<title>HTML Table Tag Usage</title>
</head>
<body>
<table border = "1">
<tr>
<th>Name</th>
<th>Country</th>
</tr>
<tr>
<td>Dhoni</td>
<td>India</td>
</tr>
<tr>
<td>David Miller</td>
<td>South Africa</td>
</tr>
<tr>
<td>Joe Root</td>
<td>England</td>
</tr>
</table>
</body>
```

Name	Country
Dhoni	India
David Miller	South Africa
Joe Root	England

Table Tags & Attributes

- ***<caption>* -** caption for the table
- ***cellspacing*** - specifies the space between table cells.
- ***cellpadding*** - specifies the space between the border of a table **cell** and its contents.
- ***Column and Row Span Attributes* -** The two or more table rows / columns can be merged into a single row / column.
- ***bgcolor*** - background of the table
- ***Height and Width of the Table* -** height and width of the table can be set by using width and height attributes
- ***allign* -** provides the alignment of content inside an element

HTML 5 - Audio

- The ***"audio"*** tag is an inline element that is used to embed sound files into a web page.

```
<audio>
  <source src="sample.mp3" type="audio/mpeg">
</audio>
```

- **Attributes:** The various attributes that can be used with the "audio" tag are listed below:
  - ***Controls:*** Designates what controls to display with the audio player.
  - ***Autoplay:*** Designates that the audio file will play immediately after it loads controls.
  - ***Loop***: Designates that the audio file should continuously repeat.
  - ***src:*** Designates the URL of the audio file.
  - ***muted:*** Designates that the audio file should be muted.

HTML 5 - Video

- The HTML5 "***video***" element specifies a standard way to embed a video in a web page.
- There are three different formats that are commonly supported by web browsers – mp4, Ogg and WebM.

```
<video src="" controls>&lt/video>
```

- **Attributes that can be used with the "video" tag are listed below :**
  - **Autoplay**
  - **Preload**
  - **Loop**
  - **Height and Width**
  - **Controls**
  - **Muted**